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# Single Nucleotide Polymorphisms and Tendon/Ligament Injuries in Athletes: A Systematic Review and Meta-analysis

Authors
**Yumi Fukuyama[1, 2], Haruka Murakami[2], Motoyuki Iemitsu[2]**

Affiliations
1 Department of Physical Therapy, Aino University, Ibaraki, Japan
2 Faculty of Sport and Health Science, Ritsumeikan University, Kusatsu, Japan

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Correspondence
Dr. Motoyuki Iemitsu
Ritsumeikan University
Faculty of Sport and Health Science
1-1-1 Nojihigashi
525-8577 Kusatsu
Japan
Tel.: + 81-77-561-3760
iemitsu@fc.ritsumei.ac.jp

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**ABSTRACT**

This systematic review and meta-analysis aimed to identify the association between genetic polymorphisms and tendon and ligament injuries in adolescent and adult athletes of multiple competition sports. The PubMed, Web of Science, EBSCO, Cochrane Library, and MEDLINE databases were searched until July 7, 2023. Eligible articles included genetic studies on tendon and ligament injuries and comparisons between injured and non-injured athletes. This review included 31 articles, comprising 1,687 injury cases and 2,227 controls, from a meta-analysis of 12 articles. We identified 144 candidate gene polymorphisms (only single nucleotide polymorphisms were identified). The meta-analyses included vascular endothelial growth factor A (VEGFA) rs699947, collagen type I alpha 1 rs1800012, collagen type V alpha 1 rs12722, and matrix metalloproteinase 3 rs679620. The VEGFA rs699947 polymorphism showed a lower risk of injuries in athletes with the C allele ([C vs. A]: OR = 0.80, 95 % CI: 0.65–0.98, $I^2$ = 3.82 %, $p$ = 0.03). The risk of these injuries were not affected by other polymorphisms. In conclusion, the VEGFA rs699947 polymorphism is associated with the risk of tendon and ligament injuries in athletes. This study provides insights into genetic variations that contribute to our understanding of the risk factors for such injuries in athletes.

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## Introduction

Athletes train to support their performance, and most will engage in activities to prevent the risk of injury. However, athletes are constantly at risk of injury. Among the sports injuries reported in high school and college athletes in the United States, tendon and ligament injuries occur at a rate of 20–30 % [1]. Tendon and ligament injuries include the Achilles tendon injury, Achilles tendinopathy, carpal tunnel syndrome, anterior cruciate ligament (ACL) injury, medial collateral ligament (MCL) injury, and rotator cuff tendinopathy. Athletes subjected to these injuries are forced to leave the competition for prolonged periods and these injuries have a significant impact on subsequent athlete performance [2, 3]. In fact, athletes who sustain an Achilles tendon injury or ACL injury return to their pre-injury condition approximately 55–80 % of the time, although not all athletes recover [4–7]. In recent years in adolescents and adults, including athletes, the number of tendon and ligament injuries has increased by approximately 20–40 % [8–10]. A reduction in such injuries is an important focus of research.

Tendon and ligament structures are composed of dense connective tissues. Although they differ in anatomical location and functional role, both tissues have similar basic components and molecular structures. The main structural component of tendons and ligaments is collagen, which accounts for 70–80 % of their dry weight, and their fibers are composed of collagen types I, III, V, VI, XII, and XIV [11, 12]. Collagen type I is the major component of collagen fibers in tendons and ligaments. Type III collagen contributes to fibrillogenesis, while type V collagen is involved in regulating the diameter of type I collagen fibrils [13, 14]. Type XII collagen is re-

sponsible for the formation of interfibrillar connections and binding to proteoglycans and decorin [15]. Each collagen fiber is related to collagen fibrillogenesis and tensile strength [16]. Other components of tendons and ligaments include decorin, aggrecan, biglycan, lumican, and other proteoglycans such as elastin, other glycoproteins, and tenascin C [12]. Decorin and biglycan regulate the assembly of collagen fibrils [17–19]. Aggrecan has a role in increasing the compression stiffness of the collagen network [20]. Elastin is a highly extensible matrix protein [20]. Tenascin C is present at the musculotendinous and osteotendinous junctions and it transmits mechanical loading [21]. Therefore, these extracellular matrices play functional roles in tendons and ligaments, such as regulated collagen fiber assembly and compression stiffness. In addition, matrix metalloproteinases (MMPs) are responsible for the breakdown of tendon and ligament structures, such as collagen and proteoglycans [22, 23]. Thus they play an important role in remodeling and support the maintenance of tendon and ligament structures. In recent studies, differences in genetic sequences encoding proteins related to tendon and ligament structures have been associated with the incidence of sports-induced tendon and ligament injuries [24]. Collagen and proteoglycan gene polymorphisms increase the fragility of collagen connective tissue and are a potential risk factor for tendon and ligament injuries [25, 26]. MMP-related gene polymorphisms may cause increased degradation of collagen and extracellular matrix (ECM) [27]. Additionally, interleukins and angiogenic endothelial growth factors are involved in tendon and ligament turnover; thus, their encoded gene polymorphisms have received attention in research [28]. Consequently, genetic variations may affect tendon and ligament injuries in athletes. In fact, the risk of injury in individuals with a family member with tendon and ligament injuries is twofold higher than that in individuals whose family members are unaffected by these injuries [29, 30].

Previous studies on sports injuries have not included integrated analyses of multiple genetic factors and tendon and ligament injuries, owing to limited sample sizes, differences in athletic competitions, and sex differences. Furthermore, previous case-control studies have been conducted, whereas the populations were often inactive, with few reports specifically comparing athletes. As a result, this study aimed to provide a systematic review and meta-analysis of the genetic associations between tendon and ligament injuries in athletes. This systematic review and meta-analysis is the first to investigate the potential effects of genetic variations on tendon and ligament injuries in athletes.

## Materials and Methods

This meta-analysis was conducted according to the 2020 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement guidelines (PRISMA checklist) [31]. The study was registered with the University Hospital Medical Information Network (UMIN #000050485).

### Literature Search and Characteristics of Studies

A literature search for systematic reviews and meta-analyses was conducted using the following five databases: PubMed, Web of Science, EBSCO, Cochrane Library, and MEDLINE. All articles were published before July 7, 2023. The search criteria were as follows: (genotype OR gene polymorphism OR SNP OR single nucleotide polymorphism OR polymorphism OR genetic variant OR allele OR genetic polymorphism) and (ligament injury OR ligament rupture OR ligament tear). The following terms were included for tendon injury: (genotype OR gene polymorphism OR SNP OR single nucleotide polymorphism OR polymorphism OR genetic variant OR allele OR genetic polymorphism) and (tendon injury OR tendinopathy OR tendon pathology OR rotator cuff injury OR rotator cuff rupture OR rotator cuff damage OR rotator cuff tear). The following terms were included for soft tissue injury: (genotype OR gene polymorphism OR SNP OR single nucleotide polymorphism OR polymorphism OR genetic variant OR allele OR genetic polymorphism) AND (soft tissue injury OR soft tissue rupture OR soft tissue tear OR soft tissue damage). The results of three searches were merged to exclude duplicates. In addition, review literature was screened, and additional articles were manually searched.

The eligibility criteria were as follows: (i) participants were athletes or players with similar activity levels; (ii) control participants were athletes with similar activity levels in the injury group; (iii) tendon and ligament injury reports were included in the results; and (iv) any injury was identified by a doctor's examination, imaging, or diagnostic report of the athlete or team staff. Limitations of the study design were not considered. Studies were excluded based on the following criteria: (i) animal studies, case reports, or case series; (ii) participants were not athletes (including recreational athletes); (iii) participants were diagnosed with a disease; or (iv) articles showing sports performance. The authors independently screened the titles, abstracts, and full texts according to the eligibility and exclusion criteria. Disagreements were resolved through discussion. In the meta-analysis, three or more articles were associated with tendon and ligament injury risk; data for the same SNP was collected from these articles. Odds ratios (OR) and 95 % confidence intervals (CIs) were calculated based on the number of participants per genotype in both groups.

### Data extraction and methodological quality assessment

Data were extracted by independent reviewers. Data were extracted using a predetermined data collection sheet that included the primary author, journal of publication, year of publication, country of origin, ethnicity of the participants, athletic competition, number of participants in the injury and control groups, genotyping method used in each study, and genotype counts of the injury and control groups.

The Newcastle-Ottawa Scale (NOS) for the assessment of non-randomized controlled trial (RCT) studies was employed to assess the methodological quality of eligible studies [32]. A score of 0–9 was assigned to each study, with a score of ≥ 6 (depicted as an asterisk) indicating that the study was of high quality. The methodological quality of each study was individually evaluated by two reviewers, and the research articles were discussed when a consensus could not be reached.

### Statistical analysis

OR and 95 % CI associated with tendon and ligament injuries were calculated using the number of participants in the injury and con-

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trol groups for each genotype from the articles related to collagen type I alpha 1 (COL1A1) rs1800012, collagen type V alpha 1 (COL5A1) rs12722, MMP3 rs679620, and vascular endothelial growth factor A (VEGFA) rs699947 polymorphisms that were deemed fit for the meta-analysis. When the number of participants for each genotype was not reported in detail, and percentages were provided, calculations were performed by the authors. For the COL1A1 rs1800012 polymorphism, the OR was calculated solely using the dominant model [GG vs. GT + TT] and allele models [G vs. T] because participants with the TT genotype were not identified in the studies by Shukla et al. [33] and Rodas G et al. [34]. For COL5A1 rs12722, MMP3 rs679620, and VEGFA rs699947 polymorphisms, a meta-analysis was performed on dominant [wild-type /wild-type (WW) vs. wild-type /variant (WV) + variant /variant (VV)], recessive [VV vs. WW + WV], and allele models [W vs. V] to determine the most appropriate genetic model related to tendon and ligament injuries risk and to examine the magnitude of genetic effects.

Calculations were performed using the DerSimonian and Laird random-effects model based on a previous study [35]. Statistical heterogeneity between studies was evaluated using the Cochran Q-statistical and $I^2$ tests ( > 50 % was considered evidence of a significant discrepancy). A statistical Z–test was used, and a $p$–value of less than 0.05 was considered to indicate statistical significance. Funnel plots and Egger regression tests were used to estimate publication bias. All statistical analyses were performed using the Stata 17.0 (Stata version 17.0; StataCorp LP, USA).

# Results

## Literature search and study characteristics

A total of 1,429 articles were identified from PubMed, Web of Science, EBSCO, Cochrane Library, and MEDLINE databases. The initial literature search identified 652 relevant articles from PubMed, 124 from the Web of Science, 294 from EBSCO, 23 from the Cochrane Library, and 336 from MEDLINE. After removing 735 duplicate records, 466 articles were excluded from the 694 articles that were ineligible based on title and abstract screening. The full texts of 228 potentially relevant articles were read to confirm the inclusion criteria, and 197 were excluded because they did not meet the eligibility criteria. Finally, 31 articles were included in this systematic review (▶ **Table 1**). The 31 articles included were published in English, and their years of publication range from 2013 to 2023. The reported injury types include ACL, MCL, Achilles tendinopathy, patellar tendinopathy, rotator cuff tendinopathy, and hip abductor tendinopathy. The competition athletes targeted in this study were soccer [24, 34, 36–49], volleyball [50–52], endurance running [53], football [54], and multisport athletes (handball, floorball, ice hockey, and basketball) [33, 55–63]. The study designs of all included articles comprised 23 case-control studies [24, 33, 37, 38, 40–46, 49–52, 55–62], five cross-sectional studies [36, 39, 47, 48, 53], two cohort studies [34, 63], and one longitudinal study [54].

This review identified 144 gene polymorphisms as candidates from 31 articles, and the meta-analysis included the COL1A1 rs1800012, COL5A1 rs12722, MMP3 rs679620, and VEGFA rs699947 polymorphisms. In this systematic review, eight articles examined the effects of the COL1A1 rs1800012 polymorphism. We did not receive data from three articles that did not provide data on the effects of the COL1A1 rs1800012 polymorphism. Ultimately, five articles were selected for this meta-analysis [24, 33, 34, 39, 63]. The effects of the COL5A1 rs12722 polymorphism did not have data available from one article; ultimately, five articles were selected for this meta-analysis [34, 39, 40, 55, 63]. Regarding articles that examined the effects of MMP3 rs679620 and VEGFA rs699947 polymorphisms, four articles each were identified and included in the meta-analysis [34, 39, 41, 51, 56, 62]. The literature screening process is shown in ▶ **Fig. 1**. Because multiple target gene polymorphisms were included in a previous study, 12 articles comprising 1,687 cases and 2,227 controls were examined using a meta-analysis (▶ **Table 2**). Four studies were conducted in Poland [24, 40, 41, 43], two in Spain [34, 55] and India [33, 62], and the remaining studies were conducted in Brazil [51] and Croatia [56], while other studies were conducted in various other countries [39, 63]. All included studies were case-control or cross-sectional, except for the cohort studies by Rodas et al. [34] and Sivertsen et al. [63]. In all studies, the genotype distribution was consistent with the Hardy-Weinberg equilibrium (HWE). In ▶ **Table 2**, COL1A1, COL5A1, MMP3, and VEGFA polymorphisms are presented as wild type/wild type (WW), wild type/variant (WV), and variant/variant (VV), respectively.

## Quality assessment

The NOS for non-RCTs was used to assess the risk of bias in the included studies (▶ **Table 3**). Nine articles received two stars for control factor items, because they were adjusted for confounders, such as age, ethnicity, sports, and level of competitive activity [24, 33, 34, 39–41, 43, 62, 63]. All articles received a total of 6–9 stars, indicating excellent quality.

## Meta-analysis

The TT homozygote of the COL1A1 rs1800012 polymorphism was not identified [24, 33, 34], and the estimated ORs were calculated using dominant [GG vs. GT + TT] and allele models [G vs. T]. The dominant ([GG vs. GT + TT]: $p = 0.95$) and allele models ([G vs. T]: $p = 0.99$) showed no significant differences in the risk of tendon and ligament injuries (see **Supplementary Fig. S1** online).

COL5A1 rs12722 and MMP3 rs679620 polymorphisms were estimated using dominant, recessive, and allele models. None of the models for COL5A1 rs12722 polymorphism showed significant effects (dominant model [CC vs. CT + TT]: $p = 0.50$; recessive model [CC + CT vs. TT]: $p = 0.88$; allele model [C vs. T]: $p = 0.60$). Similarly, no significant differences were observed for MMP3 rs679620 polymorphism in any model (dominant model [GG vs. GA + AA]: $p = 0.84$; recessive model [GG + GA vs. AA]: $p = 0.89$, and allele model [G vs. A]: $p = 0.88$) (see **Supplementary Fig. S2, S3** online).

The VEGFA rs699947 polymorphism was analyzed using dominant, recessive, and allele models. VEGFA rs699947 polymorphism showed no significant differences between the dominant ([CC vs. CA + AA]: $p = 0.19$) and recessive models ([CC + CA vs. AA]: $p = 0.16$), whereas a significant difference was observed in the allele model (C vs. A: OR = 0.80, 95 % CI: 0.65–0.98, $I^2 = 3.82$ %, $p = 0.03$). The results indicated that athletes with the C allele had a 20 % lower risk of tendon and ligament injuries than those with the A allele (▶ **Fig. 2**).

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▶ **Table 1** Summary of review articles (n = 31).

| Author | Year | Study design | Country | Sports | Diagnostic Criteria | Diagnosis | Participants | Age (years) | Gene | Polymorphism | Finding |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Alakhdar Y et al. [55] | 2023 | case-control study | Spain | Multi sports | Ultrasound | RCT | n = 137<br>Case: 49 (Male: Female/ 25:24)<br>Control: 88 (Male: Female/ 52:36) | Case: 25.2 ± 6.6<br>Control: 21.9 ± 4.4 | COL5A1<br>COL11A1<br>COL11A2 | rs12722<br>rs3753841<br>rs1676486<br>rs1799907 | The COL5A1 rs12722 CC genotype was significantly associated with the rotator cuff tendinopathy group and bilateral ultrasound pathological images ($p < 0.05$). No significant differences were found for the COL11A1 and COL11A2 gene polymorphisms. |
| Artells R et al. [36] | 2016 | cross-sectional study | Spain | Soccer | MRI<br>Ultrasound | MCL | n = 60 (Male only) | 25.52 ± 6.6 | EMILIN1 | rs2289360 | The EMILIN1 rs2289360 AA and AG genotypes were related to ligament injury rate and severity. EMILIN1 rs2289360 GG genotype players sustained no MCL injury during the 7-seasons. |
| Briški N et al. [56] | 2021 | case-control study | Croatia | Multi sports | Clinical examination | Achilles tendinopathy | n = 155<br>Case: 63 (Male: Female/ 47:16)<br>Control: 92 (Male: Female/ 72:20) | Case: 32.1 ± 12.8<br>Control: 39.0 ± 11.4 | MMP3 | rs591058<br>rs650108<br>rs679620 | The MMP3 rs650108 GG genotype ($p = 0.0074$) and rs679620 AA genotype ($p = 0.0119$) were significantly more frequent in the tendinopathy group and were associated with a predisposition to tendinopathy. |
| Cięszczyk P et al. [37] | 2017 | case-control study | Poland | Soccer | Surgery | ACL | n = 372<br>Case: 229 (Male: Female/ 158:71)<br>Control: 143 (Male: Female/ 99:44) | Case: Male 26 ± 4/ Female 25 ± 4<br>Control: Male 25 ± 3/ Female 29 ± 2 | ACAN<br>BGN<br>DCN<br>VEGFA | rs1516797<br>rs1042103<br>rs1126499<br>rs516115<br>rs699947 | The ACAN rs1516797 GT genotype (OR = 1.68, 95 % CI: 1.09–2.57, $p = 0.017$) and BGN rs1042103 A allele (OR = 1.5, 95 % CI: 1.05–2.15, $p = 0.029$) showed an increased risk of ACL injury. The DCN rs516115 AA genotype was related to the ACL injury (OR = 0.42, 95 % CI: 0.19–0.93, $p = 0.029$), whereas the AG genotype reduced the risk of ACL injury (OR = 0.45, 95 % CI: 0.21–0.98, $p = 0.042$). |
| Ficek K et al. [24] | 2013 | case-control study | Poland | Soccer | Surgery | ACL | n = 234 (Male only)<br>Case: 91<br>Control: 143 | Case: 23.0 ± 3.0<br>Control: 25.2 ± 2.6 | COL1A1 | rs1800012<br>rs1107946 | The COL1A1 rs1800012 TT genotype was less in the ACL injury group and may potentially be to protective factor in the context of ACL injury ($p = 0.084$). |
| Ficek K et al. [38] | 2014 | case-control study | Poland | Soccer | Surgery | ACL | n = 234 (Male only)<br>Case: 91<br>Control: 143 | Case: 23.0 ± 3.0<br>Control: 25.2 ± 2.6 | COL12A1 | rs970547 | No association was detected between the ACL injury in the COL12A1 rs970547 polymorphism. |
| Gutiérrez-Hellín J et al. [53] | 2021 | cross-sectional study | Spain | Endurance running | MRI<br>Ultrasound<br>Radiological examination<br>Clinical examination | Tendinopathy<br>Ligament injury | n = 89 (Male: Female/ 48:41) | range, > 20 ~ 45 | ACTN3 | rs1815739 | The ACTN3 rs1815739 RR genotype runners demonstrated a higher proportion of injuries located in the Achilles tendon ($p = 0.025$). However, the ACTN3 gene polymorphisms did not affect the type of tendon and ligament injuries. |



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▶ **Table 1** Continued.

| Author | Year | Study design | Country | Sports | Diagnostic Criteria | Diagnosis | Participants | Age (years) | Gene | Polymorphism | Finding |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Hall ECR et al. [39] | 2022 | cross-sectional study | England<br>Spain<br>Uruguay<br>Brazil | Soccer | Clinical examination<br>Medical injury reports | Tendinopathy<br>Ligament injury | n = 402 (Male only)<br>pre- peak height velocity (PHV): 101<br>post- peak height velocity (PHV): 301 | pre- PHV: 11.5 ± 1.1<br>post- PHV: 17.5 ± 2.1 | ACTN3<br>CCL2<br>COL1A1<br>COL5A1<br>EMILIN1<br>IL6<br>MMP3<br>MYLK<br>VEGFA | rs1815739<br>rs2857656<br>rs1800012<br>rs12722<br>rs2289360<br>rs1800795<br>rs679620<br>rs28497577<br>rs2010963 | In pre-peak height velocity of maturity, the COL5A1 rs12722 CC genotype was 1.7- fold more likely to be related to injury than the CT genotype ($p = 0.029$). Also, the VEGFA rs2010963 CC genotype was 10.3-fold more likely to be related to injury than the GG genotype and 11.7-fold more likely to be related to injury than the GC genotype ($p = 0.010$). |
| Jacob Y et al. [54] | 2022 | longitudinal study | Australia | Football | Medical injury reports | Tendinopathy<br>Ligament injury | n = 46 (Male only) | 21.93 ± 2.38 ~ 24.94 ± 4.26 | ACTN3<br>CCL2<br>COL1A1<br>COL5A1<br>COL12A1<br>EMILIN1<br>IGF2<br>NOGGIN<br>SMAD6 | rs1815739<br>rs2857656<br>rs1800012<br>rs12722<br>rs970547<br>rs2289360<br>rs3213221<br>rs1372857<br>rs2053423 | The IGF2 rs3213221 CC genotype was significantly associated with a higher number of tendinopathies. Additionally, a higher number of total ligament ($p = 0.019$) and non-contact ligament ($p = 0.002$) injuries were significantly associated with the COL1A1 rs1800012 TT genotype. |
| Lopes LR et al. [57] | 2021 | case-control study | Brazil | Multi sports | MRI | Patellar tendinopathy<br>RCT<br>Achilles tendinopathy | 270<br>Case: 135 (Male: Female/ 83:52)<br>Control: 135 (Male: Female/ 83:52) | range, 18 ~ 45 | TNF-α | rs1799964<br>rs1799724<br>rs1800629 | The TNF-α rs1800629 AA genotype was significantly associated with tendinopathy compared to GG + GA genotypes ($p = 0.04$). No associations were found for the TNF-α rs1799964 and rs1799724 polymorphisms. |
| Lopes LR et al. [58] | 2023 | case-control study | Brazil | Multi sports | MRI | Tendinopathy | 242<br>Case: 55 (Male: Female/ 34:21)<br>Control: 187 (Male: Female/ 124:63) | Case: 27.62 ± 6.10<br>Control: 22.90 ± 4.98 | COL1A1<br>COL1A2 | rs1107946<br>rs412777<br>rs42524<br>rs2621215 | The COL1A2 rs42524 CC and rs2621215 GG genotypes were associated with increased risk (approximately 5.5 and 4-fold, respectively) of tendinopathy. |
| Lulińska-Kuklik E et al. [40] | 2018 | case-control study | Poland | Soccer | Surgery | ACL | 345 (Male only)<br>Case: 134<br>Control: 211 | Case: 23.4 ± 3.1<br>Control: 25.3 ± 3.4 | COL5A1 | rs12722<br>rs13946 | The COL5A1 rs13946 CC + CT genotypes showed a reduced risk of ACL injury ($p = 0.039$). The CC + CT genotypes may play a protective role in ACL injury. |
| Lulińska-Kuklik E et al. [41] | 2019 | case-control study | Poland | Soccer | Surgery | ACL | 412<br>Case: 222 (Male: Female/ 156:66)<br>Control: 190 (Male: Female/ 107:83) | Case: Male 26 ± 4/ Female 25 ± 4<br>Control: Male 25 ± 3/ Female 29 ± 2 | VEGFA | rs699947<br>rs1570360<br>rs2010963 | No association was found between the VEGFA rs699947 and rs1570360 polymorphisms to ACL injury. There was a significant association between the VEGFA rs2010963 CC genotype increased risk of ACL injury ($p = 0.017$). |



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▶ **Table 1** Continued.

| Author | Year | Study design | Country | Sports | Diagnostic Criteria | Diagnosis | Participants | Age (years) | Gene | Polymorphism | Finding |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Lulińska-Kuklik E et al. [42] | 2019 | case-control study | Poland | Soccer | Surgery | ACL | 421<br>Case: 229 (Male: Female/ 164:65)<br>Control: 192 (Male: Female/ 107:85) | Case: Male 26 ± 4/ Female 25 ± 4<br>Control: Male 25 ± 3/ Female 29 ± 2 | TNC | rs1330363<br>rs2104772<br>rs13321 | No association was found between the ACL injury in the TNC gene polymorphisms. |
| Lulińska-Kuklik E et al. [43] | 2019 | case-control study | Poland | Soccer | Surgery | ACL | 421<br>Case: 229 (Male: Female/ 158:71)<br>Control: 192 (Male: Female/ 107:85) | Case: Male 26 ± 4/ Female 26 ± 6<br>Control: Male 25 ± 3/ Female 29 ± 2 | MMP3<br>MMP8<br>TIMP2 | rs591058<br>rs679620<br>rs11225395<br>rs4789932 | The MMP3 rs679620 G and rs591058 C alleles were both significantly associated with ACL injury (OR = 1.38, 95 % CI: 1.05–1.81, $p = 0.021$). |
| Lulińska-Kuklik E et al. [44] | 2019 | case-control study | Poland | Soccer | Surgery | ACL | 423<br>Case: 229 (Male: Female/ 164:65)<br>Control: 194 (Male: Female/ 109:85) | Case: Male 26 ± 4/ Female 25 ± 4<br>Control: Male 25 ± 3/ Female 29 ± 2 | IL1B<br>IL6<br>IL6R | rs16944<br>rs1143627<br>rs1800795<br>rs2228145 | The IL6 rs1800795 GG + CG genotypes were more significantly associated compared to the CC genotype to the ACL injury (OR = 1.74, 95 % CI: 1.08–2.81, $p = 0.032$). |
| Lulińska-Kuklik E et al. [45] | 2020 | case-control study | Poland | Soccer | Surgery | ACL | 430<br>Case: 228 (Male: Female/ 157:71)<br>Control: 202 (Male: Female/ 117:85) | Case: Male 26 ± 4/ Female 25 ± 4<br>Control: Male 26 ± 6/ Female 29 ± 2 | MMP1<br>MMP10<br>MMP12 | rs1799750<br>rs486055<br>rs2276109 | No association with ACL injury was found for all genes. |
| Lulinska E et al. [46] | 2023 | case-control study | Poland | Soccer | Surgery | ACL | 421<br>Case: 229 (Male: Female/ 164:65)<br>Control: 192 (Male: Female/ 107:85) | Case: Male 26 ± 4/ Female 26 ± 6<br>Control: Male 26 ± 6/ Female 29 ± 2 | ELN<br>FMOD | rs2071307<br>rs7543148 | The ELN rs2071307 AA genotype was significantly more frequent in the ACL group and was associated with a predisposition to ACL injury ($p = 0.00109$). The FMOD rs7543148 TT genotype was significantly less in the ACL group and associated with a protective role in ACL injury ($p = 0.03919$). |
| Mirghaderi SP et al. [59] | 2022 | case-control study | Iran | Soccer<br>Volleyball<br>Basketball<br>Handball | Clinical examination<br>MRI | ACL | 400 (Male only)<br>Case: 200<br>Control: 200 | Case: 30.3 ± 6.4<br>Control: 29.7 ± 6.0 | COL1A1 | rs1107946 | No association with ACL injury was found for the COL1A1 rs1107946 polymorphism. |
| Perini JA et al. [60] | 2022 | case-control study | Brazil | Multi sports | Clinical examination<br>MRI | ACL | 338<br>Case: 146 (Male: Female/ 93:53)<br>Control: 192 (Male: Female/ 129:63) | Case: 27.2 ± 6.1<br>Control: 23.0 ± 5.0 | COL1A1<br>COL1A2 | rs1107946<br>rs412777<br>rs42524<br>rs2621215 | The frequency of COL1A2 rs2621215 G allele was higher in non-contact ACL injury group than in the control group and was associated with an increased risk of non-contact ACL injury (OR = 1.69, 95 % CI: 1.02–2.81, $p = 0.04$). No association with ACL injury was found for other gene polymorphisms. |



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▶ **Table 1** Continued.

| Author | Year | Study design | Country | Sports | Diagnostic Criteria | Diagnosis | Participants | Age (years) | Gene | Polymorphism | Finding |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Pruna R et al. [47] | 2013 | cross-sectional study | Spain | Soccer | MRI Ultrasound | MCL | 73 (Male only) | 26.2 (range, 19–35) | EMILIN1 TTN SOX15 IGF2 CCL2 TNC COL1A1 COL5A1 | rs2289360 rs27423237 rs4227 rs3213221 rs2857656 rs21104772 rs1800012 rs12722 | The ligament injury associated with the EMILIN1 rs2289360 AA genotype was more severe than those associated with the AG or GG genotypes ($p = 0.009$). In comparison to the EMILIN1 rs2289360 GG (37.5 days) or AA (83.2 days) genotypes, ligament injury related to the AG genotype needed a shorter mean recovery time (24.7 days) ($p = 0.043$). |
| Pruna R et al. [48] | 2015 | cross-sectional study | Spain | Soccer | MRI Ultrasound | MCL | 73 (Male only) | 26.2 (range, 19–35) | EMILIN1 TTN SOX15 IGF2 CCL2 TNC COL1A1 COL5A1 | rs2289360 rs27423237 rs4227 rs3213221 rs2857656 rs21104772 rs1800012 rs12722 | The EMILIN1 rs2289360 GG genotype resulted in 1 mild injury (100 %), while the AG + AA genotypes resulted in 4 moderate injuries (80 %) and 1 severe injury (20 %) ($p = 0.05$). Additionally, the IGF2 rs3213221 GG genotype produced 1 severe injury (100 %), while the GC + CC genotypes produced 1 mild injury (20 %) and 4 moderate injuries (80 %), indicating a difference in injury patterns ($p = 0.05$). |
| Rodas G et al. [61] | 2019 | case-control study | Spain | Multi sports | Physical examination Ultrasound MRI | Tendinopathy | 363 Case: 199 (Male: Female/ 174:25) Control: 164 (Male: Female/ 149:15) | Case: 25 ± 7 Control: 25 ± 6 | GJA1 VAT1L CNTNAP2 and other genes | rs11154027 rs4362400 rs10263021 and 55 SNPs | The GJA1 rs11154027 1 + A allele (OR = 2.11, 95 % CI: 1.07–4.19, $p = 1.01 \times 10\text{-}6$) and the VAT1L rs4362400 G allele (OR = 1.98, 95 % CI: 1.05–3.73, $p = 9.6 \times 10\text{-}6$) were associated with a high risk to tendinopathy. In contrast, possession of the CNTNAP2 rs10263021 1 + A allele may play a protective role in tendinopathy (OR = 0.42, 95 % CI: 0.20-0.91, $p = 4.5 \times 10\text{-}6$). |
| Rodas G et al. [34] | 2023 | cohort study | Spain | Soccer | Clinical examination | ACL | 46 (Male: Female/ 22:24) | Not shown | COL5A1 and other genes | rs13946 rs16399 rs1134170 rs71746744 rs3196378 and 103 SNPs | The COL5A1 rs13946 CC genotype has a significant association with ACL injury in female athletes ($p = 0.017$). Also, the COL5A1 rs3196378 A allele showed a significant association with ACL injury ($p = 0.040$). No associations were found in male athletes. |



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▶ **Table 1** Continued.

| Author | Year | Study design | Country | Sports | Diagnostic Criteria | Diagnosis | Participants | Age (years) | Gene | Polymorphism | Finding |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Salles JI et al. [50] | 2015 | case-control study | Brazil | Volleyball | MRI | Patellar tendinopathy<br>RCT Achilles tendinopathy<br>Hip abductors tendinopathy | 138 (Male only)<br>Case: 52<br>Control: 86 | Case: 30.23 ± 4.72<br>Control: 27.33 ± 4.67 | BMP4<br>FGF3<br>FGF10<br>FGFR1 | rs2761884<br>rs17563<br>rs2855529<br>rs2071047<br>rs762642<br>rs7932320<br>rs1893047<br>rs12574452<br>rs4631909<br>rs4980700<br>rs1448037<br>rs900379<br>rs1011814<br>rs593307<br>rs13317 | Athletes with the BMP4 rs2761884 GT + TT genotypes were more likely to develop tendinopathy (OR = 2.39, 95 % CI: 1.10–5.19, $p = 0.01$). Additionally, the BMP4 rs2761884 T allele was highly associated with tendinopathy (OR = 2.01, 95 % CI: 1.16–3.48, $p = 0.007$). The FGF3 rs12574452 AG + GG genotypes were associated with the occurrence of tendinopathy (OR = 5.23, 95 % CI: 0.24–13.2, $p = 0.08$). |
| Salles JI et al. [51] | 2016 | case-control study | Brazil | Volleyball | MRI | Patellar tendinopathy<br>RCT Achilles tendinopathy | 179<br>Case: 88 (Male: Female/ 59:29)<br>Control: 91 (Male: Female/ 39:52) | Case: 23.0 ± 4.1<br>Control: 20.2 ± 5.1 | VEGFA<br>KDR | rs699947<br>rs833061<br>rs3025039<br>rs2071559<br>rs2305948<br>rs1870377 | Athletes with the KDR rs2305948 GA and GA + AA genotypes were less in the tendinopathy group compared to the control group and demonstrated a reduced risk of tendinopathy (OR = 0.41, 95 % CI: 0.19–0.88, $p = 0.02$ and OR = 0.47, 95 % CI: 0.23–0.98, $p = 0.04$). |
| Salles JI et al. [52] | 2018 | case-control study | Brazil | Volleyball | MRI | Patellar tendinopathy<br>RCT Achilles tendinopathy | 271<br>Case: 146 (Male: Female/ 117:29)<br>Control: 125 (Male: Female/ 73:52) | Case: 26.86 ± 6.03<br>Control: 21.62 ± 5.39 | FCRL3<br>FOXP3 | rs7528684<br>rs3761549 | The tendinopathy group presented significantly increased FCRL3 rs7528684 C allele frequency compared to the control group (OR = 1.44, 95 % CI: 1.02–2.04, $p = 0.04$). |
| Shukla M et al. [33] | 2020 | case-control study | India | Multi sports | Radiological examination Surgery | ACL | 166<br>Case: 90 (Male: Female/ 75:15)<br>Control: 76 (Male: Female/ 59:17) | Case: Male 27.2 ± 6.5/ Female 23.4 ± 2.9<br>Control: Male 24.7 ± 4.5/ Female 22.6 ± 1.8 | CO-L1A1 | rs1800012 | No significant difference was detected between the ACL and control groups in the COLIA1 rs1800012 GT or TT genotypes. |



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▶ **Table 1** Continued.

| Author | Year | Study design | Country | Sports | Diagnostic Criteria | Diagnosis | Participants | Age (years) | Gene | Polymorphism | Finding |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Shukla M et al. [62] | 2020 | case-control study | India | Multi sports | Radiological examination Surgery | ACL | 166 Case: 90 (Male: Female/ 75:15) Control: 76 (Male: Female/ 59:17) | Case: Male 27.2 ± 6.5/ Female 23.4 ± 2.9 Control: Male 24.7 ± 4.5/ Female 22.6 ± 1.8 | VEGFA | rs699947 rs35569394 | The VEGFA rs699947 A allele was significantly more prevalent in the ACL group (C vs. A allele: OR = 1.68, 95 % CI: 1.08–2.60, $p = 0.021$, and CC vs. CA + AA: OR = 2.69, 95 % CI: 1.37–5.26, $p = 0$ .004). The VEGFA rs35569394 I allele was significantly more in the ACL group (D vs. I allele: OR = 1.64, 95 % CI: 1.06–2.55, $p = 0.025$ and DD vs. ID + II: OR = 2.61, 95 % CI: 1.31–5.21, $p = 0.006$). The VEGFA rs699947 A allele and VEGFA rs35569394 I allele were associated with a predisposition to ACL injury. |
| Sivertsen EA et al. [63] | 2019 | cohort study | Norway Finland | Multi sports | MRI Arthroscopic examination | ACL | 851 (Female only) Case: 119 Control: 732 | 19.9 ± 4.4 | COL1A1 COL3A1 COL5A1 COL12A1 | rs1800012 rs1107946 rs1800255 rs12722 rs13946 rs970547 | No association with ACL injury was found for all gene polymorphisms. |
| Sun Z et al. [49] | 2022 | case-control study | Poland | Soccer | Surgery | ACL | 430 Case: 228 (Male: Female/ 157:71) Control: 202 (Male: Female/ 117:85) | Case: Male 26 ± 4/ Female 26 ± 6 Control: Male 26 ± 6/ Female 29 ± 2 | COL22A1 | rs11784270 rs6577958 | No association with ACL injury was found for all gene polymorphisms. |

RCT: Rotator Cuff Tendinopathy, MCL: Medial collateral ligament ACL: Anterior cruciate ligament.



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▶ **Table 2** Main characteristics of included studies and gene polymorphisms in cases and controls of meta-analysis articles.

| No. | Study | Country | Sports | Match | Diagnosis | Gene polymorphism | case | | | control | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | WW | WV | VV | WW | WV | VV |
| 1 | Ficek K et al. [24] | Poland | Soccer | ethnicity, competition exposure time | ACL injury | COL1A1 rs1800012 | 65 | 26 | 0 | 96 | 41 | 6 |
| 2 | Hall ECR et al. [39] | England | Soccer | competition | Tendinopathy Ligament injury | COL1A1 rs1800012 | 46 | 19 | 5 | 217 | 103 | 12 |
| | | Spain | | | | COL5A1 rs12722 | 19 | 30 | 21 | 70 | 148 | 114 |
| | | Uruguay | | | | MMP3 rs679620 | 15 | 41 | 14 | 95 | 163 | 74 |
| | | Brazil | | | | | | | | | | |
| 3 | Rodas G et al. [34] | Spain | Soccer | competition | ACL injury | COL1A1 rs1800012 | 6 | 2 | 0 | 25 | 13 | 0 |
| | | | | | | COL5A1 rs12722 | 4 | 1 | 3 | 9 | 19 | 10 |
| | | | | | | MMP3 rs679620 | 2 | 5 | 1 | 11 | 18 | 9 |
| | | | | | | VEGFA rs699947 | 3 | 3 | 2 | 12 | 18 | 8 |
| 4 | Shukla M et al. [33] | India | Multi sports | age | ACL injury | COL1A1 rs1800012 | 76 | 14 | 0 | 64 | 12 | 0 |
| 5 | Sivertsen EA et al. [63] | Norway | Multi sports | | ACL injury | COL1A1 rs1800012 | 79 | 38 | 2 | 512 | 201 | 20 |
| | | Finland | | | | COL5A1 rs12722 | 16 | 55 | 48 | 124 | 351 | 257 |
| 6 | Lulińska-Kuklik E et al. [40] | Poland | Soccer | age, sex, ethnicity, exposure time, competition | ACL injury | COL5A1 rs12722 | 23 | 66 | 45 | 42 | 107 | 62 |
| 7 | Alakhdar Y et al. [55] | Spain | Multi sports | age | Rotator Cuff Tendinopathy | COL5A1 rs12722 | 7 | 31 | 11 | 5 | 47 | 36 |
| 8 | Lulińska-Kuklik E et al. [43] | Poland | Soccer | ethnicity, exposure time, competition | ACL injury | MMP3 rs679620 | 68 | 107 | 54 | 40 | 93 | 59 |
| 9 | Briški N et al. [56] | Croatia | Multi sports | ethnicity | Achilles Tendinopathy | MMP3 rs679620 | 27 | 30 | 16 | 32 | 51 | 9 |
| 10 | Salles JI et al. [51] | Brazil | Volleyball | age, sex, practice time, competition | Patellar Tendinopathy Achilles Tendinopathy Rotator Cuff Tendinopathy | VEGFA rs699947 | 37 | 43 | 8 | 36 | 48 | 7 |
| 11 | Lulińska-Kuklik E et al. [41] | Poland | Soccer | ethnicity, competition, exposure time | ACL injury | VEGFA rs699947 | 62 | 121 | 39 | 66 | 99 | 25 |
| 12 | Shukla M et al. [62] | India | Multi sports | age | ACL injury | VEGFA rs699947 | 20 | 51 | 19 | 33 | 30 | 13 |

ACL: Anterior cruciate ligament.



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▶ **Table 3** Quality assessment of included studies in the meta-analysis.

| No. | Study | Selection | | | | Control for important factor or additional factor | Exposure | | | Total |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Adequate definition of cases | Representative-ness of cases | Selection of control subjects | Definition of control subjects | | Exposure assess-ment | Same method of ascertainment for all subjects | Non-re-sponse rate | |
| 1 | Ficek K et al. [24] | * | * | * | * | ** | * | * | * | 9 |
| 2 | Hall ECR et al. [39] | * | * | * | * | ** | - | * | * | 8 |
| 3 | Rodas G et al. [34] | * | * | * | * | ** | - | * | * | 8 |
| 4 | Shukla M et al. [33] | * | * | * | * | ** | * | * | * | 9 |
| 5 | Sivertsen EA et al. [63] | * | * | - | * | ** | - | * | * | 7 |
| 6 | Lulińska-Kuklik E et al. [63] | * | * | * | * | ** | * | * | * | 9 |
| 7 | Alakhdar Y et al. [55] | * | - | * | * | * | * | * | - | 6 |
| 8 | Lulińska-Kuklik E et al. [43] | * | * | * | * | ** | * | * | * | 9 |
| 9 | Briški N et al. [56] | * | * | - | * | * | - | * | * | 6 |
| 10 | Salles JI et al. [51] | * | * | * | * | * | * | * | * | 8 |
| 11 | Lulińska-Kuklik E et al. [41] | * | * | * | * | ** | * | * | * | 9 |
| 12 | Shukla M et al. [62] | * | * | * | * | ** | - | * | * | 8 |

Funnel plots did not show obvious asymmetry, suggesting no significant publication bias for VEGFA polymorphisms (▶ **Fig. 3**). Other gene polymorphisms showed no significant publication bias. Furthermore, Egger's test revealed no statistically significant publication bias (**see Supplementary Fig. S4–S6** online). A subgroup analysis could not be performed because of the small number of articles included in the meta-analysis.

## Discussion

To the best of our knowledge, this is the first study to investigate genetic factors associated with tendon and ligament injuries in athletes. This review selected 31 articles that reported genetic factors associated with tendon and ligament injuries in athletes, and 22 genes were found to be associated with the risk of tendon and ligament injuries. In this meta-analysis, the VEGFA rs699947 (−2578 C/A) C allele was associated with a 20 % lower risk of tendon and ligament injury. Thus this polymorphism may reduce the risk of tendon and ligament injuries in athletes.

All the genetic polymorphisms identified in this study were SNPs. SNPs affect the expression of mRNA and proteins by replacing the base sequences of protein-coding and non-coding regions of DNA with different bases. The SNPs associated with tendon and ligament injuries in athletes identified in this study are involved in the structure and metabolic turnover of tendons and ligaments and may result in individual differences in the occurrence of injuries. Collagen comprises approximately 75 % of the dry weight of tendons and ligaments, with type I collagen accounting for 85 % of this weight [12, 64]. Type I collagen is a heterotrimer consisting of two α1 chains and one α2 chain encoded by COL1A1 and COL1A2 genes [65, 66]. The rs1800012 ( + 1245 G/T) polymorphism identified in the first intron region of COL1A1 is associated with mRNA expression in tendons and ligaments [67]. A combined meta-analysis of five articles suggested no significant risk of tendon and ligament injuries for COL1A1 rs1800012 polymorphism [24, 33, 34, 39, 63]. Wang et al. [68], in a meta-analysis based on five articles that included 933 cases with 1,381 controls, demonstrated a significant difference in the risk of tendon and ligament injuries between individuals with the TT and GG + GT genotypes, including non-athletes. Since the meta-analysis in this study did not include individuals with the TT genotype in two articles, we could not examine the effect of the TT genotype on these injuries [24, 33, 34]. For this reason, additional studies are needed to determine the association between the COL1A1 rs1800012 polymorphism and tendon and ligament injuries, particularly in athletes. The COL1A1 rs1800012 polymorphism is located at the Sp1 binding site and upregulates COL1A1 mRNA transcription by substituting G with T. The mRNA ratio of α1 to α2 in type I collagen is approximately 2:1 [67]. However, the composition ratio differed depending on genotype. As type I collagen contributes to the tensile strength of tendons and ligaments, the COL1A1 rs1800012 polymorphism may be related to the development of tendon and ligament injuries. However, further studies are needed to examine the relationship between COL1A1 polymorphisms and tendon and ligament injuries in athletes.

Type III collagen is a major component of tendons and ligaments, and is involved in the structure of type I collagen fibrils

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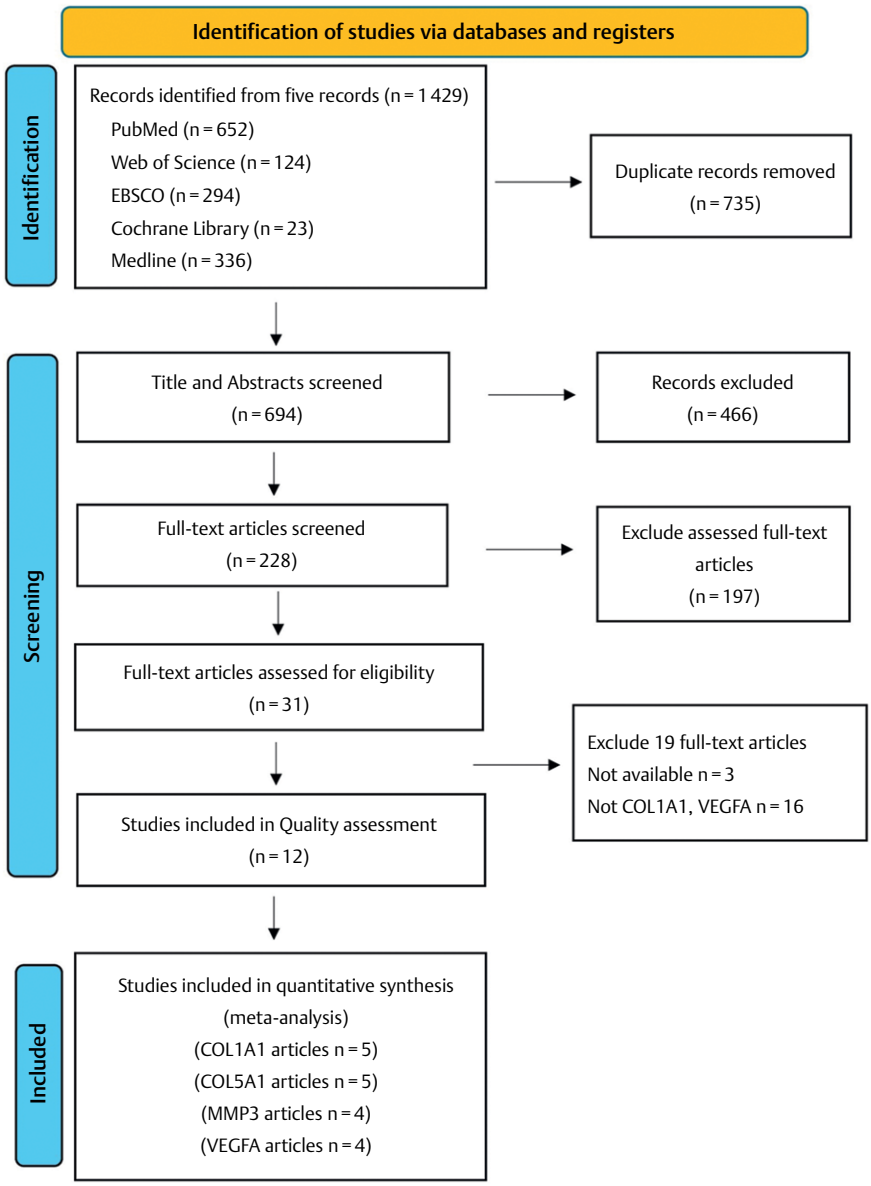


▶ **Fig. 1** Summary flowchart of literature search.

[69, 70]. Additionally, type III collagen is regulated during the wound response process and is increased in tendinopathy [71]. The rs1800255 (2209 G/A) polymorphism within exon 30 of the COL3A1 gene, which codes for the α1(III) chain of type III collagen, causes the amino acid substitution from alanine to threonine by substituting G by A [72]. Threonine has larger and more hydrophilic side chains than alanine, which may disrupt the triple-helical structure of type III collagen. This base substitution affects the assembly and tensile strength of collagen fibers, and may be related to tendon and ligament injuries [73]. Sivertsen et al. [63] found no significant association between the COL3A1 rs1800255 polymorphism and ACL injury in athletes. However, nonathletic research suggests a higher risk of incident ligament injuries with the AA genotype than with the GG + GA genotype [26, 74]. Owing to the paucity of athletic studies included in this review, it is difficult to demonstrate the effect of the COL3A1 rs1800255 polymorphism.

Type V collagen, which accounts for approximately 10 % of the collagen components of tendons and ligaments, regulates the diameter of type I collagen fibers [13, 75]. Type V collagen α1 chain is encoded by the COL5A1 gene. The COL5A1 rs12722 (414 C/T) and rs13946 (230 C/T) polymorphisms were identified to be located within the 3′-untranslated regions (3′UTR). The COL5A1 polymorphism leads to a 50 % decrease in type V collagen and causes poor connective tissue organization, resulting in impaired tensile strength [76]. Lulińska-Kuklik et al. [40] found that the frequencies of the COL5A1 rs13946 CC and CT genotypes were higher in a control group than in an ACL group of soccer players. Rodas et al. [34] showed a significant association between the CC genotype and ACL



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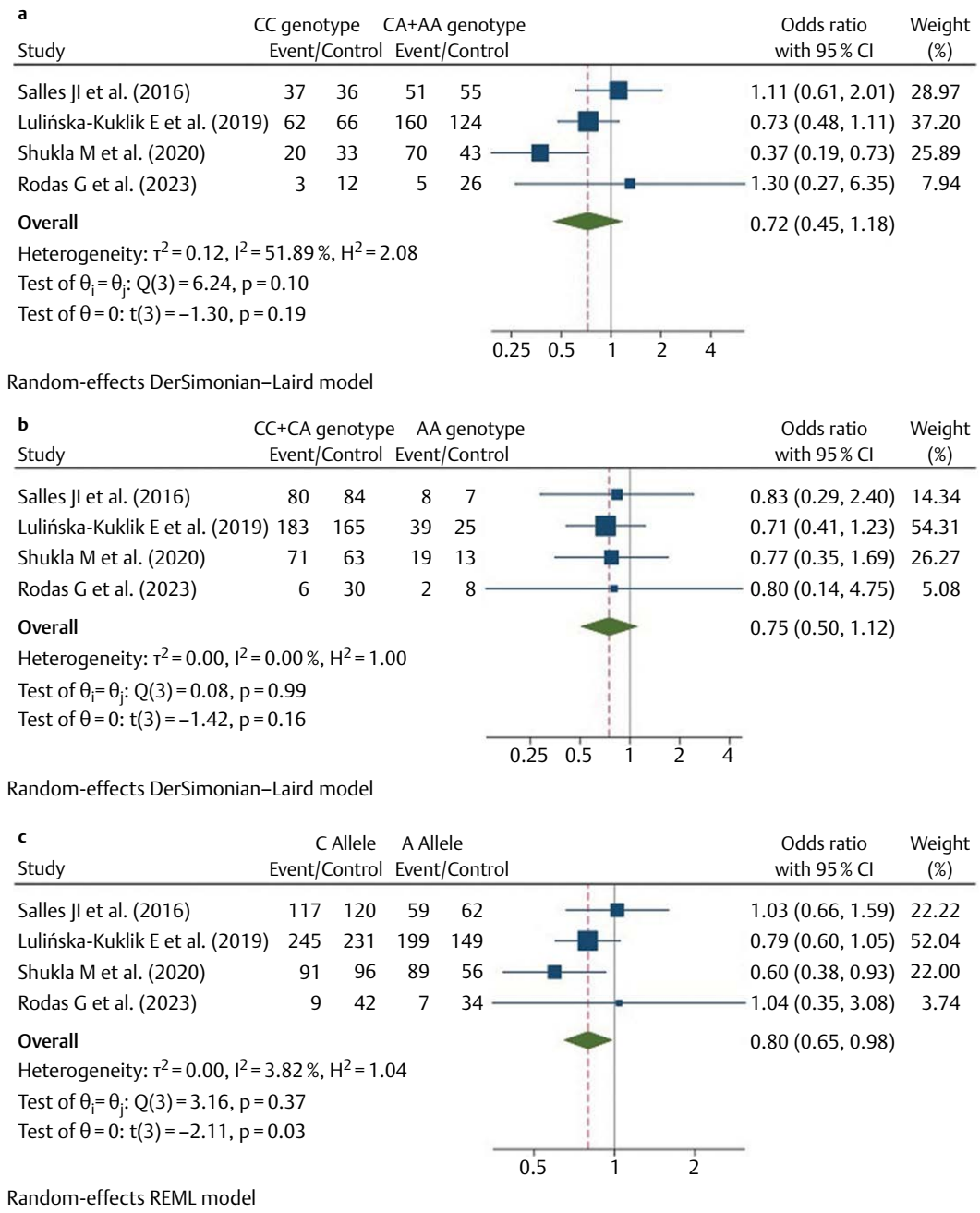


▶ **Fig. 2** Forest plots of dominant (**a**), recessive (**b**), and allele models (**c**) of VEGF rs699947 for tendon and ligament injuries in all articles. Squares and horizontal lines correlate to ORs and 95 % CIs for individual studies. The area of the squares indicates the weights of the studies. The diamonds indicate the combined ORs and 95 % CIs. Event = injury, Control = non-injury.

injuries in female athletes. Additionally, Alakhdar et al. [55] and Hall et al. [39] showed that the frequency of COL5A1 rs12722 CC genotype was higher in tendon and ligament injury groups than in the control group. However, Sivertsen et al. [63] showed that ACL injury in multi-sport female athletes was not affected by the COL5A1 rs12722 and rs13946 polymorphisms. Our meta-analysis also showed a lack of association between COL5A1 rs12722 polymorphism and tendon and ligament injuries. To clarify this discrepancy, further studies are required to examine the effects of COL5A-1gene polymorphisms in matched competitive sports.

Type XII collagen consists of a homotrimer of three α1(XII) chains encoded by the COL12A1 gene and plays a role in the interactions between extracellular molecules and the organization of collagen fibril bundles [15, 77]. The rs970547 (9285 A/G) polymorphism, located in exon 65 of the COL12A1 gene, causes an amino acid substitution from serine to glycine by substituting A for G. In

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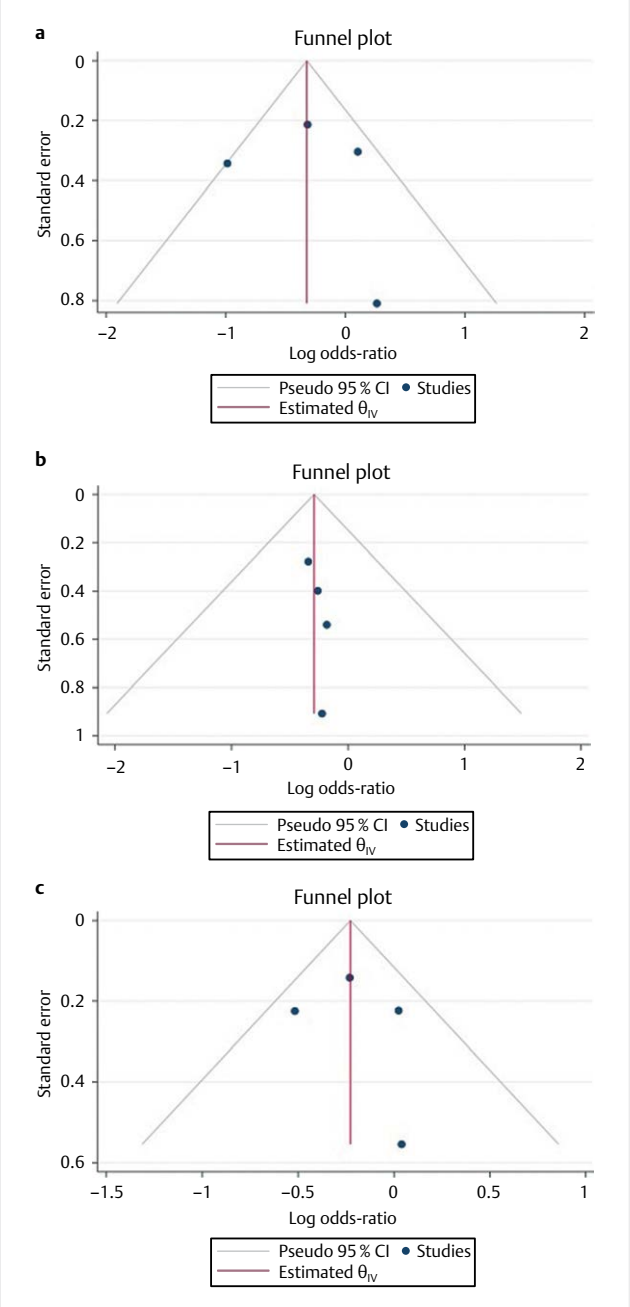


▶ **Fig. 3** Funnel plot analysis for publication bias of VEGF rs699947 for tendon and ligament injuries in all articles.

this review, Ficek et al. [38] and Sivertsen et al. [63] demonstrated the lack of an effect of the rs970547 polymorphism on ACL injury in athletes. These findings may be related to the limited sample size and the recruitment of athletes from multiple sports.

MMP are one of the 23 types of MMPs expressed in humans. Its primary role involves ECM degradation and remodeling, contributing to the metabolic turnover of various collagens and glycoproteins [78, 79]. The MMP family includes the collagenases MMP1 and MMP8, stromelysins MMP3 and MMP10, and macrophage metalloelastase MMP12, which are located at loci on chromosome 11q22.3 [78, 80]. The tissue inhibitor of matrix metalloproteinases (TIMP) family comprises four members (TIMP1, TIMP2, TIMP3, and TIMP4), and they inhibit a majority of MMPs by forming non-covalent binary complexes [81, 82]. TIMP-2 regulates collagen ECM remodeling via MMPs activation [83, 84].

MMP3 is involved in the degradation of aggrecan, elastin, fibronectin, and laminin in various extracellular matrix proteins, and collagen types II, III, IV, V, IX, and X [78, 80]. The identified MMP3 gene polymorphisms include the rs591058 polymorphism, in which T is substituted by C and is located in intron 4; the rs650108 polymorphism, in which G is substituted by A and is located in intron 8; and the rs679620 (276 G/A) polymorphism, which leads to the conversion of lysine to glutamic acid by the substitution of A by G and is located in exon 2. Briški et al. [56] observed that Achilles tendinopathy was significantly associated with the rs679620 AA and rs650108 GG genotypes. Additionally, Lulińska-Kuklik et al. [43] showed that MMP3 rs679620 G and rs591058 C alleles were associated with ACL injury. However, our meta-analysis using four articles found no association with tendon and ligament injuries [34, 39, 43, 56]. This discrepancy in SNP effects may be because of the large sample sizes reported by Lulińska-Kuklik et al. [43] and Hall et al. [39], whereas other studies had smaller sample sizes. Because small sample sizes affect meta-analysis statistics, inconsistencies may arise. Therefore, the effects of MMP3 polymorphisms on tendon and ligament injuries need to be verified in a large cohort.

Other MMP polymorphisms may also be associated with tendon and ligament injuries in athletes. However, the MMP1 rs1799750 (−1607 1 G/2 G), MMP8 rs11225395 (−799 C/T), MMP10 rs486055(180 A/G), and MMP12 rs2276109 (−82 A/G) polymorphisms are not associated with tendon and ligament injuries in athletes [43, 45]. Additionally, the TIMP2 rs4789932 (−2803 T/C) polymorphism, which regulates MMP activity, did not affect ACL injury in athletes [43]. However, there is only one study on tendon and ligament injuries associated with each of these polymorphisms in athletes. Further studies are required to examine the effects of these polymorphisms.

Genes related to angiogenesis, interleukins, and fibroblasts have been found to be associated with inflammation [85]. Early inflammation induces the expression of inflammatory cytokines and growth factors, including interleukins [86–88]. Subsequently, fibroblasts and angiogenesis are induced and tendon repair occurs through proliferation and remodeling, contributing to the maintenance of tendon homeostasis [86, 87]. Angiogenic signaling is regulated by vascular endothelial growth factors (VEGF). VEGF in the A isoform (VEGFA) has the highest angiogenic capacity, and VEGFA protein expression is increased during the repair of injured tendons and ligaments [89, 90]. This causes an increase in MMP and decrease in TIMP3 expression via VEGFA expression [91–93]. Thus the tendon and ligament structural components of collagen and ECM are degraded, resulting in the locked integrity of the structure and an increased likelihood of injury [92, 94, 95]. The VEGFA rs699947 (−2578 C/A) C allele identified in the promoter region of the VEGFA gene is associated with increased VEGFA protein expression [96]. In four articles combined to perform a meta-analysis of 408 cases and 395 controls, the VEGFA rs699947 polymorphism was found to affect tendon and ligament injuries [34, 41, 51, 62]. The meta-analysis of this review suggests that there is a 20 % lower risk of tendon and ligament injuries in the athletes with VEGFA rs699947 C

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allele. This discrepancy in SNP effects may be more susceptible to the results of a single study because of the small number of reported studies and low statistical power. Therefore, the effect of the VEGFA rs699947 polymorphism on tendon and ligament injuries needs to be verified in a large cohort.

In other VEGFA gene polymorphisms, Lulińska-Kuklik et al. [41] and Hall et al. [39] showed that the VEGFA rs2010963 ( − 634 G/C) CC genotype located in the promoter increases the risk of tendon and ligament injuries in athletes. The VEGFA rs2010963 CC genotype increases VEGFA protein expression and may consequently affect these injuries [97]. Furthermore, Shukla et al. [62] confirmed significantly more carriers of the VEGFA rs35569394 ( − 2549 I/D [Insertion/Deletion]) I allele, which is located in the promoter region in the athletes within the ACL injury group. Thus gene polymorphisms other than VEGFA rs699947 may affect these injuries in athletes; however, further studies are required.

Additionally, regarding other inflammatory response-related gene polymorphisms, Lulińska-Kuklik et al. [44] showed that the interleukin-6 (IL-6) rs1800795 ( − 174 G/C) G allele increases the risk of ACL injury in athletes. The IL-6 rs1800795 G allele increases IL-6 production by promoting IL-6 mRNA transcription [98–100] and the expression of VEGF due to increased IL-6 release may affect tendon and ligament turnover, resulting in an increased risk of ACL injury [101–103]. Furthermore, the fibroblast growth factor 3 (FGF3) rs12574452 AG + GG genotype, which promotes collagen synthesis in tendons and ligaments, and bone morphogenetic protein 4 (BMP4) rs2761884 GT + TT genotype, which mutually regulates cartilage and tendon differentiation, are associated with an increased risk of tendon and ligament injuries [50]. Although the glycoprotein Fc receptor 3 (FCRL3) rs7528684 ( − 169 C/T) C allele, a member of the immunoglobulin receptor superfamily, increases the risk of tendon and ligament injuries [52], the effects of these polymorphisms remain unclear. Thus some polymorphisms in inflammatory response-related genes may influence tendon and ligament injuries. However, further studies are needed to clarify the effects of these genetic polymorphisms.

The ECM contains glycoproteins, such as elastin, proteoglycan, fibronectin, and tenascin-C. Elastin is responsible for 1–2 % of the dry weight of tendons, which contributes to the elasticity of tendons and ligament fibers and is crucial for external pressures [64, 104–107]. The elastin microfibril interface-located protein (EMILIN1) binds to elastin and microfibrils and plays a role in stabilizing microfibrils. Proteoglycans have excellent affinity and are involved in the resistance of collagen fibers to compressive and tensile strengths [108]. Aggrecan (ACAN) binds to type I collagen to stabilize the collagen network [108]. Biglycan (BGN) and decorin (DCN) are involved in collagen fibrillogenesis and modulate fiber formation and structure [108, 109]. Additionally, tenascin-C (TNC) proteins are involved in signal transduction processes such as cell adhesion and cell proliferation/migration [21, 110]. The EMILIN1 rs2289360 A allele, ACAN rs1516797 GT genotype, and BGN rs1042103 A allele are associated with an increased risk of tendon and ligament injuries [36, 37, 48]. In contrast, the DCN rs516115 AG genotype was associated with a reduced risk of tendon and ligament injuries [37]. Furthermore, the TNC gene polymorphisms showed no significant associations. ACAN, BGN, DCN, and EMILIN1 polymorphisms are located in non-coding regions that may affect mRNA stability between genotypes [111, 112]. However, these four gene polymorphisms may not only affect ECM composition whereas also collagen fiber tensile strength and compliance. Therefore, ECM-related gene polymorphisms may increase or reduce the risk of tendon and ligament injuries.

A genome-wide association study (GWAS) of gene polymorphisms associated with tendinopathy identified three gene polymorphisms, i. e. the gap junction alpha 1 (GJA1) rs11154027, vesicular amine transport 1-like (VAT1L) rs4362400, and contactin-associated protein-like 2 (CNTNAP2) rs10263021 [61]. Furthermore, a 7-year longitudinal study showed that the insulin-like growth factor-2 (IGF2) rs3213221 CC genotype is associated with non-contact tendinopathy [54]. However, the mechanisms by which these genetic polymorphisms play a role remain unclear.

The present study has several limitations regarding genetic influences on tendon and ligament injuries in athletes. First, the meta-analysis of the VEGFA rs699947 polymorphism included only a few previous studies on athletes (four articles). Although other genetic polymorphisms have been reported, few studies fulfill the criteria for a meta-analysis. Further investigation is needed to enhance the quality of the evidence. Second, the mechanisms of some of the SNPs, such as COL12A1 and MMPs, are still unclear, and their effects on tendon and ligament injuries need to be clarified. Third, many articles in this study focused on the athletic population in the Nordic region, and it was difficult to compare the genetic influences with other ethnic groups. Finally, it may be necessary to examine acquired effects, such as the athlete's competitive level, daily training time, and nutrition. These results suggest that understanding the genetic profiles of individual athletes may be the first step toward injury prevention. It is important to continue investigating the effects of genetic factors in larger sample sizes and different ethnic groups to develop more explicit strategies for biological profiling.

In conclusion, this systematic review and meta-analysis reveals that genetic factors affect the risk of tendon and ligament injuries in athletes. Furthermore, a meta-analysis showed that the VEGFA rs699947 C allele is associated with a reduced risk of tendon and ligament injuries in athletes. However, heterogeneity was confirmed among studies on the risk of developing tendon and ligament injuries in athletes, and only four articles reported the VEGFA rs699947 polymorphism. For this reason, our results partially explained the effects of genetic polymorphisms on tendon and ligament injuries in athletes. Further studies are needed to determine the genetic profiles of athletes with tendon and ligament injuries.

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## Conflict of Interest

There is no conflict of interest for any author.

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